



Chapter 15 — Multi-Operator Collaboration & War-Room Mode



Feature summary: Multiple users join the **same live run** with role-based consoles — the commander sees the full picture, operators see their assigned sectors — with shared presence (live cursors), a commander **annotation layer** (arrows, zones, notes drawn on the map for everyone), and structured task assignment. This turns DOIP from a single-user dashboard into a **command center that a team operates together**.

15.1 Concept & Motivation

Real operations centers are staffed by teams, not individuals. War-Room Mode makes DOIP's training story real:

- **Training exercises** become multi-trainee drills with an instructor commanding.
- **Coordination itself** becomes a trainable skill: handoffs, task assignment, escalation discipline.
- **The demo impact doubles:** two laptops side by side, one commander drawing an arrow, one operator watching it appear — instantly communicates “real-time collaborative platform.”

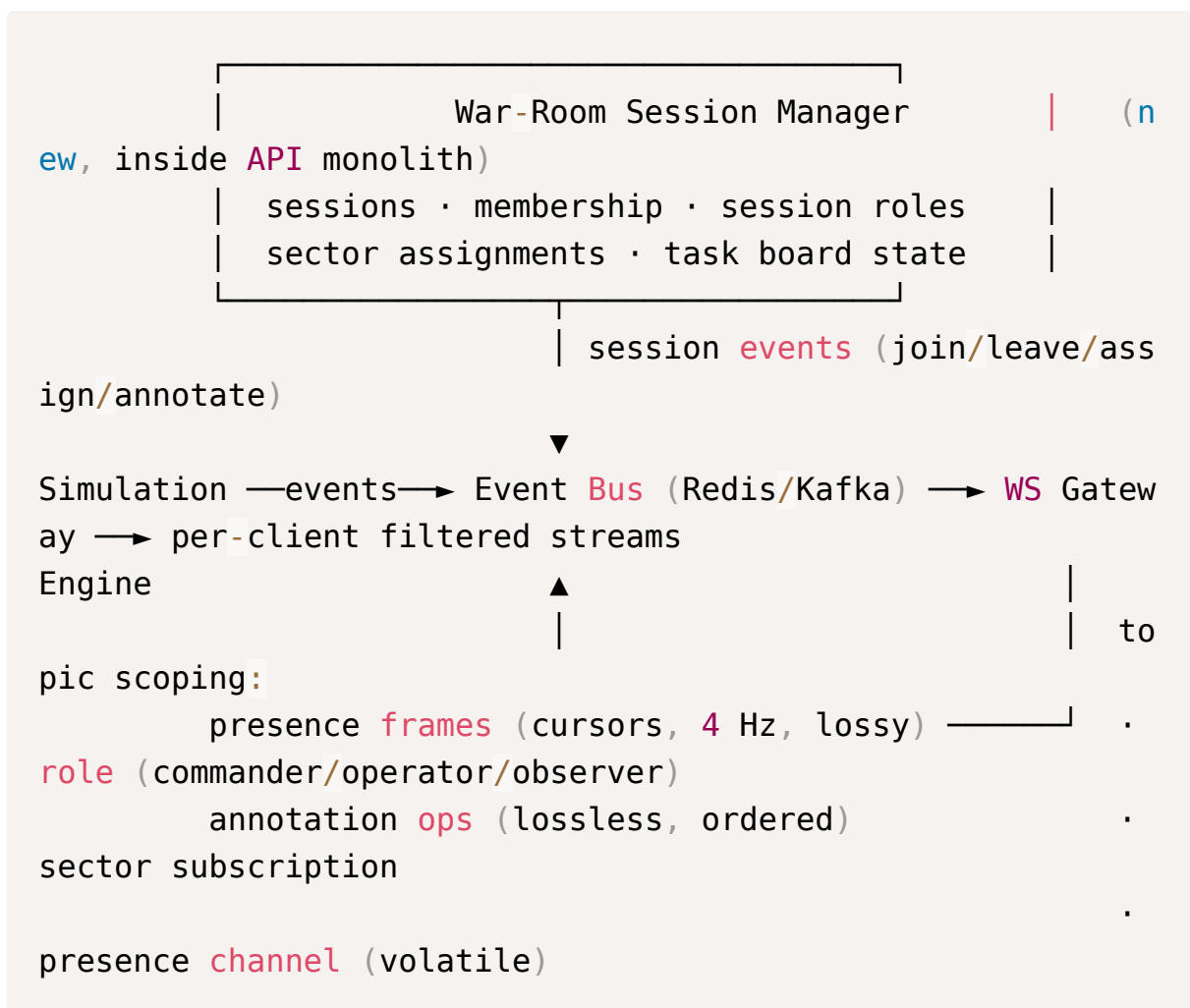
Engineering honesty: ~70% of the machinery (WebSocket fan-out, event bus, RBAC, run lifecycle) already exists. What's new is **session semantics, presence, annotations, and scoped views**.

15.2 Roles in a War-Room Session

Session role	Sees	Can do
Commander (1 per session)	Everything, all sectors	Draw/edit annotations, assign tasks, control run (pause/speed), broadcast alerts
Sector Operator (N)	Assigned sector(s) + global critical events	Acknowledge/resolve incidents in sector, complete tasks, request assistance
Observer / Instructor (N)	Everything, read-only + scoring tools	Add private evaluation notes, trigger scenario injects (if also admin)

Session roles are *per-run* and layered on top of platform RBAC (Ch. 10): you must be at least `operator` to take a session role; only `planner` + can be commander.

15.3 Architecture



Key design decisions

1. Annotations and tasks are events, not ad-hoc state.

`AnnotationCreated/Updated/Deleted`, `TaskAssigned/Accepted/Completed`, `SessionMemberJoined/Left` flow through the standard event schema (Ch. 6). Consequences: they appear in replay (a debrief shows *what the commander drew and when*), they survive reconnects via the existing resume protocol, and the audit trail is free.

2. Presence is volatile, not evented.

Live cursors and “who's looking where” go over a separate lossy presence channel (like position frames, 4 Hz, no persistence) — keeping the event log clean.

3. Last-writer-wins per annotation, no CRDTs.

Annotations are commander-owned objects with single-writer semantics; version numbers detect conflicts. Full CRDT machinery is deliberately out of scope (see risks).

4. Sector scoping is a server-side filter

in the WS gateway (never client-side hiding): operators receive only their sector's position frames + global critical events. This doubles as a performance win — each client gets a fraction of the traffic.

15.4 Data Model Additions (extends Ch. 5)

```
CREATE TABLE war_room_sessions (  
  id UUID PRIMARY KEY, run_id UUID REFERENCES runs(id),  
  name TEXT, created_by UUID REFERENCES users(id),  
  status TEXT CHECK (status IN ('open', 'active', 'debrief', 'closed')),  
  created_at TIMESTAMPTZ DEFAULT now());  
  
CREATE TABLE session_members (  
  session_id UUID REFERENCES war_room_sessions(id),  
  user_id UUID REFERENCES users(id),  
  session_role TEXT CHECK (session_role IN ('commander', 'operator', 'observer')),  
  sector_ids UUID[], -- operator sector assignments  
  joined_at TIMESTAMPTZ, PRIMARY KEY (session_id, user_id));  
  
CREATE TABLE annotations (  
  id UUID PRIMARY KEY, run_id UUID REFERENCES runs(id),  
  name TEXT, created_by UUID REFERENCES users(id),  
  status TEXT CHECK (status IN ('open', 'active', 'debrief', 'closed')),  
  created_at TIMESTAMPTZ DEFAULT now());
```

```

    id UUID PRIMARY KEY, session_id UUID REFERENCES war_room_
sessions(id),
    author_id UUID REFERENCES users(id),
    kind TEXT CHECK (kind IN ('arrow', 'polygon', 'marker', 'tex
t', 'ping')),
    geometry GEOMETRY NOT NULL,    -- PostGIS, SRID 4326
    style JSONB, label TEXT, version INT DEFAULT 1,
    created_at TIMESTAMPTZ, updated_at TIMESTAMPTZ, deleted_a
t TIMESTAMPTZ);

CREATE TABLE session_tasks (
    id UUID PRIMARY KEY, session_id UUID REFERENCES war_room_
sessions(id),
    assigned_to UUID REFERENCES users(id), incident_id UUID N
ULL,
    title TEXT, status TEXT CHECK (status IN ('assigned', 'acc
epted', 'done', 'declined')),
    created_at TIMESTAMPTZ, resolved_at TIMESTAMPTZ);

```

15.5 Functional Requirements

- FR-15.1: Create/join a war-room session for any run via invite link or session list; membership and roles visible to all members.
- FR-15.2: Live presence: member list with status, and shared map cursors (name-tagged, smoothly interpolated, auto-hidden when idle > 10 s).
- FR-15.3: Commander annotation tools: arrow, polygon, marker, text label, and a transient **ping** (3-second pulse for "look here"); all render on every member's map < 500 ms after creation.
- FR-15.4: Sector assignment UI: commander drags operators onto sectors; operators' map view auto-scopes; reassignment takes effect live.
- FR-15.5: Task board: commander assigns tasks (optionally bound to an incident); operators accept/complete; task state changes appear in the event feed and replay.
- FR-15.6: Broadcast alert: commander sends a priority banner to all members (eventuated, acknowledged per member).

- FR-15.7: Debrief mode: replay of the run shows annotations/tasks appearing at their original times; per-operator activity timeline for instructor review (feeds Ch. 11 §11.6 metrics).
- FR-15.8: Graceful degradation: a member disconnecting/reconnecting resumes with full session state (annotations, tasks, assignments) via snapshot + event resume — zero lost session events.
- FR-15.9: Session capacity target: 10 concurrent members per session, 2 concurrent sessions per instance, within existing NFR budgets.

15.6 Implementation Plan

Sprint	Work
15-A	Session manager (CRUD, membership, roles), session event types + reducers, member list UI
15-B	Presence channel (volatile cursor frames), sector-scoped WS filtering, operator scoped view
15-C	Annotation tools (MapLibre draw + event round-trip + LWW versioning), ping, broadcast alerts
15-D	Task board, debrief replay integration, instructor activity timelines, load test (10 members)

Effort: ~4 sprints (5–6 weeks). Depends on: WS resume protocol (Ph. 1), replay (Ph. 2), RBAC (Ph. 1). Recommended slot: **after Phase 5, before Phase 6** — or post-v1 as Phase 8 if solo timeline is tight.

15.7 Testing Additions

- Multi-client integration tests: two headless WS clients assert annotation propagation latency and ordering.
- Partition test: operator disconnects during 20 session events → resumes with byte-identical session state.
- Authorization matrix: operator cannot annotate, observer cannot acknowledge — parameterized over all session roles.
- Playwright E2E: two browser contexts in one test — commander draws, operator sees it.

15.8 Risks

Risk	Impact	Mitigation
Scope creep toward full collaborative editing (CRDTs, co-editing annotations)	High	Hard rule: single-writer annotations, LWW versioning only; documented as ADR
Presence traffic bloats WS load	Medium	4 Hz cap, delta suppression when idle, presence excluded from event log
Session state divergence across clients	Medium	All session state is evented + resumable; nightly partition tests
Demo requires multiple humans	Low	Scripted headless “ghost operator” client for solo demos